Answers on
Booklet - 10

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a)	**spreading out** of waves (into a 'shadow region') (WTTE)	M1	
	when the waves meet an aperture or obstacle (WTTE)	A1	[2]
(b)	**narrow gap**:		
	almost full semicircles (no straight section)		
	with gap at centre	B1	
	wide gap:		
	virtually straight (must have some straight section)		
	with gap at centre	B1	
	wavelength shown to be **constant** – generously judged by eye	B1	
	OR same λ labelled before & after aperture on either diagram		[3]

2.

(a) (i)	any 3 from reflection, TIR, refraction, diffraction, interference	B1+B1+B1	[3]
	{also allow energy transfer, superposition, creation of standing waves but do not allow $v=f\lambda$, progressive waves}		
(ii)	polarisation	B1	[1]
(b) (i)	frequency = number of vibrations/waves/oscillations/cycles per sec	B1	
	{reject f=1/period unless period defined}		
	wavelength = distance between neighbouring corresponding pts	B1	[2]
	{Allow "crest to crest" "trough to trough" or labelled diagram or length of 1 cycle of the wave}		
(c)	in one second f waves are produced each of length λ	B1	
	distance travelled in one second is therefore $f\lambda$ OR (hence) $v = f\lambda$	B1	[2]
	OR speed = dist/time = λ/T	B1	
	and $T = 1/f$ hence $v= f\lambda$	B1	
	{reject consistency of units approach}		
(d) (i)	period = 1/500 = 0.002s (or 2ms)	B1	[1]
(ii)	at least 2 full (sine) waves of constant period (+/- 2mm)	B1	
	of amplitude 3 cm (+/- 2mm in both directions)	B1	
	correct 'period' of 4 cm (+/- 2mm throughout)	B1	[3]
(iii)	correct substitution into $v=f\lambda$: e.g. $330 = 500\lambda$	C1	
	$\lambda = 0.66$ m {do not allow 0.6 but allow 0.7}	A1	[2]

3.

(a) (i) amplitude = 1.2 (mm) B1 [1]

(ii) period = 2.4 (ms) B1 [1]
{allow 2.4×10^{-3} ms if 2.4×10^{-3} is correctly used in substitution in b(i)}

(b) (i) frequency = 1/period C1
1/0.0024 = 417Hz (OR 420) A1 [2]
{1/2.4 = 0.417 OR 0.42 OR 0.4 scores 1 mark}
{allow ecf from cand's period value}

(ii) recall of $v = f\lambda$ OR $c = f\lambda$ OR $\lambda = vT$ OR $1500 = 417\lambda$ C1
$\lambda = 3.6$ m A1 [2]
{ecf for cand's f: e.g. $\lambda = 1500/0.417 = 3600$ m scores 2 marks
OR $\lambda = 1500/0.4 = 3750$ m scores 2 marks}
$\lambda = 1500/0.42 = 3571$ m scores 2 marks

(iii) valid scale for cand's λ shown on position axis AND at least two full 'sine' waves drawn (waves can be very rough but not square waves } B1

amp. shown as 1.2 mm +/- ½ sq.:check first peak + trough only B1

first wavelength correct as 3.6 m +/- 1 sq.{allow ecf from (b) (ii) } B1 [3]
{NB If there is no scale on the position axis the 1st and 3rd marks cannot be scored}

4.

(a)(i) (transmitter emits) microwaves that are (vertically plane) POLARISED B1
{allow vibrate/move/travel in the same plane, but do NOT allow 'transverse'}

detector ONLY detects waves (polarised) PARALLEL to it (WTTE) B1 [2]

(ii) maximum reading OR reading would increase (WTTE) B1 [1]

(iii) the waves are TRANSVERSE {allow polarised if not gained in (i)} B1 [1]

5.

i	UV-A causes tanning or skin ageing ; most of (99%) uv light; 400-315 nm UV-B causes damage or sunburn or skin cancer; 315-260 nm UV-C is filtered out by atmosphere/ozone layer; 260-100 nm	B1 B1 B1	accept values within ranges with tolerance of 20 nm **allow** $\lambda_A > \lambda_B > \lambda_C$ for 1 mark **max 3 marks** from 7 marking points
ii	filters out/blocks/reflects/absorbs UV(-B)	B1	**allow** chemicals prevent sunburn/skin cancer **not** stops UV penetrating skin

6.

(a)	Arrow (within the lemon and) towards the negative terminal		**B1**
(b)(i)	$\Delta Q = I\Delta t$	(Allow other subject. Δ is not necessary)	**C1**
	charge = $1.2 \times 10^{-3} \times 6.9 \times 10^{5}$		**C1**
	charge = $828 \approx 830$ (C)	(-1 for 10^{n} error and -1 for t = 8 days)	**A1**

7.

(a)(i)	Ammeter	B1
(ii)	Voltmeter	B1
(b)	I: current	B1
	Δt: time (interval / duration)	B1
(c)(i)	$I = 7.5 \times 10^{3} / 1500$	C1
	$I = 5.0$ (A)	A1

8.

(a)	$\rho = RA / L$		M1
	Symbols defined:		
	ρ = resistivity, A = cross-sectional area, R = resistance and L = length		A1
	($R = \rho L/A$ with all symbols defined scores 1/ 2)		
(b)(i)	$A = \rho L / R$		C1
	$A = 4.3 \times 10\text{-}6 \times 1.2 \times 10^{-2} / 5.0$	(-1 for using L as 2.0 mm)	C1
	$A = 1.0(3) \times 10^{-8}$ (m^2)		A1
(ii)	$t = 1.0(3) \times 10^{-8} / 2.0 \times 10^{-3}$		
	$t = 5.1(6) \times 10^{-6}$ (m)	(possible e.c.f)	B1

9.

(a)(i)	p.d.: Energy 'lost' by charge(s)/ electron(s) (as heat / light)	B1
	e.m.f: Energy 'gained' by charge(s)/ electron(s) (as electrical)	B1
	(p.d. linked to energy transfer to heat/light <u>and</u> e.m.f to energy transfer to electrical scores 1/2)	
(ii)	Tick <u>only</u> for J C^{-1}	B1

10.

(a)	R = V / I	M1
	Symbols defined: R = resistance, V = p.d. and I = current	A1
	(Allow use of 'voltage' instead of 'p.d.')	
	(Resistance = p.d. <u>per</u> (unit) current scores 2/2)	
	(V = IR with all symbols defined scores 1/2)	
	(Resistance = p.d. <u>per</u> (unit) amp / A scores 1/2)	
	(Resistance = volts <u>per</u> (unit) current scores 1/2)	
	(Resistance = volts <u>per</u> (unit) ampere scores 0/2)	
(b)(i)	$I \propto V$ / I ÷ V = constant	C1
	Current ∝ p.d. / p.d. ÷ current = constant <u>and</u> (metallic conductor at) constant temperature	A1
	(Allow symbols in (b)(i) if defined in (a))	
(ii)	One correct response scores 1/2	
	Three correct responses scores 2/2	
(c)	Ammeter in series	B1
	Voltmeter across cell or the thermistor	B1
	(No credit if the meters or their positions are contradictory)	

11.

(p.d =) energy (transfer)/charge	B1	**Allow** $(V =)\frac{W}{Q}$ or $(V =)\frac{E}{Q}$ where W (or E) = energy / work (done) and Q = charge. (1) For mark above, there is no need to define V. **Not** energy lost **by** unit charge / by coulomb **For a mark the idea of 'division' or 'per' is important**

12.

(a) All I-V graphs identified correctly B1

(b)(i) The resistance / R remains constant B1
$I \propto V$ / Graph has a constant slope / gradient / Obeys Ohm's law B1

(b)(ii) R is infinite / large when: $I = 0$ / no conduction / 'reverse' direction / 'up to a point' / negative V / negatively biased B1
R is small / decreases / low(er) when: I is not zero / there is conduction / 'positive' direction / 'beyond a point' / positive V / positively biased B1
(No credit for 'conducts in one direction only')

(b)(iii) The resistance increases as I (or V) increases B1
The temperature increases (as I increases) / more electrons collisions (with the vibrating atoms / ions) B1

13.

(a) (Semiconductor) diode B1

(b) The diode symbol circled (No ecf allowed) B1

(c) $R = \frac{V}{I}$ C1
At 0.20 V, R = infinite / very large A1
At 0.70 V, $R = (\frac{0.70}{0.020} =)35\ (\Omega)$ (Allow answers in the range: {31.82 to 38.89}) A1

14.

(a) $\rho = RA / L$ (Allow $R = \rho l/A$) M1

Symbols defined:

(ρ = resistivity) A = cross-sectional area, R = resistance and L = length A1

(resistivity = product of resistance and cross sectional area per (unit) length scores 2/2)

(resistivity = product of resistance and cross sectional area per (unit) metre scores 1/2)

(b)(i) $h = 1.2 \times 10^{-5} / 3.0 \times 10^{-4}$ C1

$h = 4.0 \times 10^{-2}$ (m) A0

(ii) $R = \rho l/A$

$R = 6.9 \times 10^{-2} \times 4.0 \times 10^{-2} / 3.0 \times 10^{-4}$ (-1 for 10^n error) C1

$R = 9.2(0)$ (Ω) A1

($R = 920$ (Ω) scores 1/2)

(c) Resistance decreases M1

by a factor of four because the length is halved (and area is doubled) A1

(Numerical approach with $R = 2.3$ (Ω) scores 2/2)